

TECHNICAL NOTE

CRIMINALISTICS/TOXICOLOGY

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Determination of Herbicides Paraquat, Glyphosate, and Aminomethylphosphonic Acid in Marijuana Samples by Capillary Electrophoresis

ABSTRACT: In this work, two methods were developed to determine herbicides paraquat, glyphosate, and aminomethylphosphonic acid (AMPA) in marijuana samples by capillary electrophoresis. For paraquat analysis, sample was extracted with aqueous acetic acid solution and analyzed by capillary zone electrophoresis with direct UV detection. The running electrolyte was 50 mmol/L phosphate buffer (pH 2.50). For glyphosate and AMPA, indirect UV/VIS detection was used, as these substances do not present chromophoric groups. Samples were extracted with 5 mmol/L hydrochloric acid. The running electrolyte was 10 mmol/L gallic acid, 6 mmol/L TRIS, and 0.1 mmol/L CTAB (pH = 4.7). The methods presented good linearity, precision, accuracy, and recovery. Paraquat was detected in 12 samples ($n = 130$), ranging from 0.01 to 25.1 mg/g. Three samples were positive for glyphosate (0.15–0.75 mg/g), and one sample presented AMPA as well. Experimental studies are suggested to evaluate the risks of these concentrations to marijuana user.

KEYWORDS: forensic science, paraquat, glyphosate, AMPA, marijuana, capillary electrophoresis, Cannabis, forensic toxicology

According to the World Drug Report 2010 issued by the United Nations Office on Drugs and Crime (1), marijuana remains as one of the most consumed drug worldwide, with almost 165.6 million users. A significant increase has been noticed in Brazil from 2006 to 2007, probably due to the greater availability of *Cannabis* products coming from neighbor countries (1,2).

Some authors showed marijuana contamination by cosmopolitan microorganisms (fungi, for example *Aspergillus*) most of them as mycotoxin producers, which present great toxicological significance (3,19,20). They can lead to occupational allergy in police officers, hypersensitive toxicologists and also in users as they have direct contact through dispersion, inhalation and/or ingestion of spores or hyphae when handling or using the drug. So far, one case of death has been reported as a consequence of chronic use of marijuana contaminated by *Aspergillus*, due to lung aspergillosis (3).

Paraquat is a quaternary ammonium compound used as nonselective contact herbicide, used as desiccant and defoliant for the control of weeds and grasses growing in fruit orchards. This

substance has a good occupational safety record (occupational injuries are associated with dermal and eyes irritation), but present a high mortality index after intentional exposure (4,5).

Herbicides have been extensive used to control drug production in last three decades. At the end of 70's, substances such as 2,4-D and paraquat were used to eradicate marijuana crops in Mexico. Such action was quite controversial due to the paraquat toxicity levels for marijuana users as well as for inhabitants near to the cultivation and herbicide application sites (6, 18). In the last decade, glyphosate [N-(phosphonomethyl)glycine] was used in coca plants eradication program in Colombia (7).

This work describes the development of an analytical procedure for determination paraquat, glyphosate and its main breakdown product—the aminomethylphosphonic acid (AMPA), in marijuana samples by capillary electrophoresis with direct and indirect detection. The proposed procedure is useful to appraisals and assessments on the possible risks that marijuana users can be exposed.

Materials and Methods

Reagents and Chemicals

Paraquat dichloride (1,1'-dimethyl-4,4'-bipyridinium dichloride hydrate), diquat dibromide monohydrate (1,1'-ethylene-2,2'-bipyridylium dibromide monohydrate), glyphosate (N-phosphonomethyl-glycine), aminomethyl-phosphonic acid (AMPA) were purchased from Aldrich Chem. Co. (Milwaukee, WI). Ethyl paraquat dibromide (1,1'-diethyl-4,4'-bipyridinium dibromide) (used as internal standard for paraquat analyses) and glutamic

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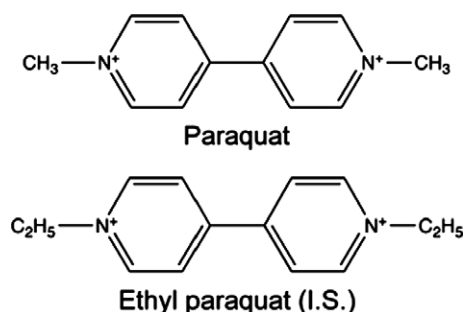


FIG. 1—Chemical structure of paraquat and ethyl paraquat (internal standard).

acid (used as internal standard for glyphosate and AMPA analyses) were also purchased from Aldrich Chem. Co. Figure 1 depicts molecular structures of paraquat and ethyl paraquat. Figures 2 and 3 depict molecular structures and acid dissociation constant of glyphosate and AMPA, respectively.

Acetic acid (99.7%), hydrochloric acid (37%), tris(hydroxymethyl)aminomethane (TRIS), and cetyl trimethylammonium (CTAB) were purchased from Merck (Darmstadt, Germany). Water was purified in a Milli-Q water system (Millipore, Billerica, MA).

To prepare paraquat's extraction solvent (5 mmol/L aqueous acetic acid), 0.287 mL of acetic acid (99.7%) was slowly transferred to a 1-L volumetric flask containing 250 mL of ultrapure water, and the volume was completed with the same solvent. To prepare glyphosate and AMPA's extraction solvent (5 mmol/L aqueous hydrochloric acid), 0.411 mL of hydrochloric acid (37% w/w) was slowly transferred to a 1-L volumetric flask containing 250 mL of ultrapure water, and the volume was completed with the same solvent.

Marijuana blank samples (without herbicides) were donated by UNODC through the International Collaborative Exercise (ICE) drug analysis program certification.

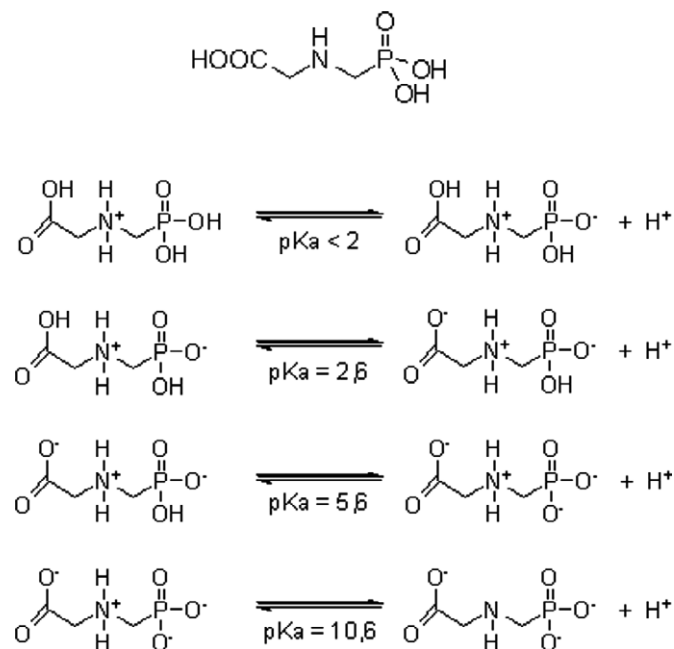


FIG. 2—Chemical structure and acid dissociation constant of glyphosate.

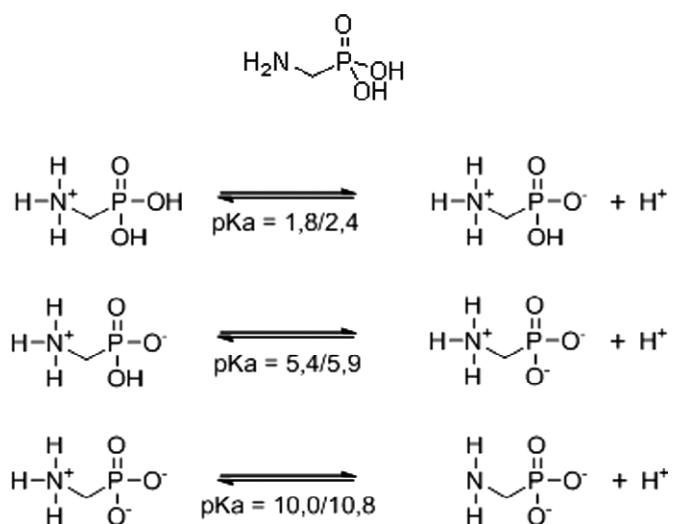


FIG. 3—Chemical structure and acid dissociation constant of AMPA.

Preparation of Standard Solutions

Stock solutions of paraquat, glyphosate, AMPA, ethyl paraquat, and glutamic acid were prepared separately by dissolving an appropriate amount of each compound with ultrapure water in volumetric glassware, to achieve a concentration of 1.0 mg/mL. Working solutions from 5 to 1000 µg/mL (six calibrators and three controls for each analyte) were prepared through serial dilution of the stock standard solutions with ultrapure water. All stocks and working solutions were stored at -5°C.

Instrumentation

Paraquat Analyses—The capillary zone electrophoresis experiments were carried out on a capillary electrophoresis system, model HP^{3D}CE (Agilent Technologies, Palo Alto, CA), equipped with on-column diode-array detection (DAD) and controlled by software HP ChemStation, version A.08.03 (Agilent Technologies, Palo Alto, CA, USA). An uncoated fused-silica capillary (Polymicro Technologies, Phoenix, AZ) with an inner diameter of 75 µm and a total length of 48.5 cm (40 cm effective length) was used. The capillary was thermostatted at 21°C. New capillaries were conditioned with 1 mol/L NaOH solution (30 min), followed by deionized water (20 min). At the beginning of the day, the capillary was conditioned with 1 mol/L NaOH for 5 min, followed by deionized water for 5 min and then run electrolyte. Between electrophoretic runs, the capillary was rinsed with run electrolyte for 3 min. Samples were introduced onto the capillary via hydrodynamic injection by applying pressure in inlet vial (50 mbar for 10 sec). Phosphate buffer 50 mmol/L (pH 2.50) was used as running electrolyte. The analyses were conducted under normal polarity (+21 kV) and direct UV detection (λ = 195 nm).

Glyphosate and AMPA Analyses—Glyphosate and AMPA analyses were performed on a Beckman P/ACE MDQ CE system (Beckman Instruments, Fullerton, CA) equipped with a UV detector. Data collection, processing, and analysis were performed using system 32 Karat 7.0 software (Beckman Instruments). An uncoated fused-silica capillary (Polymicro Technologies) with an inner diameter of 75 µm and a total length of 37 cm (30 cm effective length) was used. The capillary was thermostatted at 25°C. New capillaries were conditioned

with 1 mol/L NaOH solution (30 min), followed by deionized water (20 min). At the beginning of the day, the capillary was conditioned with 1 mol/L NaOH for 5 min, followed by deionized water for 5 min and then run electrolyte. Between electrophoretic runs, the capillary was rinsed with run electrolyte for 3 min. Samples were introduced onto the capillary via hydrodynamic injection by applying pressure in inlet vial (3 psi for 5 sec). The running electrolyte was composed by 10 mmol/L gallic acid, 6 mmol/L TRIS, and 0.1 mmol/L CTAB (pH = 4.7). The analyses were conducted under reverse polarity (−20 kV) and indirect detection (λ = 254 nm).

Sample Preparation

Paraquat Analyses—Seized marijuana samples were powdered with an electric mixer and homogenized, and 1.0 g of these materials was transferred into a 15-mL conical polypropylene tube, and 1 mL of an ethyl paraquat working solution (50 $\mu\text{g/mL}$) was added. After homogenization in a mechanical agitator, 10 mL of aqueous acetic acid (5 mmol/L) was added, the tube was closed and agitated for 5 min, rested for 1 h, agitated again for more 5 min and finally centrifuged at 450 $\times g$ for 3 min. The upper phase was filtered under pressure with a syringe through a 0.45- μm membrane (Millipore) directly into the injection vial.

Glyphosate and AMPA Analyses—Seized marijuana samples were powdered with an electric mixer and homogenized, and 1.0 g of these materials was transferred into a 15-mL conical polypropylene tube, and 1 mL of a glutamic acid working solution (500 $\mu\text{g/mL}$) was added. After homogenization in a mechanical agitator, 10 mL of 5 mmol/L aqueous hydrochloric acid was added, the tube was closed and agitated for 5 min, rested for 1 h, agitated again for more 5 min and finally centrifuged at 2000 rpm for 3 min. The upper phase was filtered under pressure with a syringe through a 0.45- μm membrane (Millipore) directly into the injection vial.

Methods Validation

The validation of methods was performed by establishing selectivity, linearity, limits of detection (LOD), limits of quantification (LOQ), intra/inter-assay precision, accuracy, recovery, and stability of the analytes. To obtain these validation data, two kinds of samples were prepared: calibration samples and quality control samples (QC). These samples were independently prepared by spiking blank marijuana samples with appropriate volumes of paraquat, glyphosate, and AMPA work solutions. QC samples were prepared at concentrations 5, 40, and 80 $\mu\text{g/g}$ (QC1, QC2, and QC3, respectively).

Specificity—This parameter was assessed through the analyses of diverse marijuana samples, that were compared to blank samples spiked with paraquat, diquat (DQ) another herbicide found in formulations together with paraquat, glyphosate, AMPA, and internal standard as to obtain a final concentration of 40 $\mu\text{g/g}$. The method was considered specific if no significant interfering was detected at the absolute migration time or relative migration time.

Linearity—The linearity was established by analyzing blank marijuana samples spiked in five replicates at the following concentrations for paraquat: 5; 25; 100; 300; 600 and 1000 $\mu\text{g/g}$. For glyphosate and AMPA, linearity was evaluated at the

concentrations: 10; 25; 100; 300; 600 and 1000 $\mu\text{g/g}$. The calibration curves were obtained by plotting the peak-area ratios (analyte/internal standard) against the concentrations, and the linearity, which was expressed as the correlation coefficient (r), was obtained. The extracts of samples with concentration above the dynamic range ($>1000 \mu\text{g/g}$) were appropriately diluted, re-analyzed to achieve the calibration range and the concentration was corrected by corresponding dilution factor.

Limit of Detection and Limit of Quantification—The empirical method was used to determine LOD and LOQ comprising the analysis of several marijuana blank samples added to decreasing quantities of the analyte at issue. The limit of detection was considered as being the smallest concentration, which presented a coefficient of variation (CV) ($n = 5$) that did not exceed 20%. The limit of quantification was the smallest concentration as well, presenting a coefficient of variation that did not exceed 10% (8).

Intra-day and Inter-day Precision—Both intra- and inter-assay precision studies were indirectly performed through imprecision calculation by determining the relative standard deviation (RSD). The intra-assay precision (repeatability) was determined through the analysis, conducted on the same day, of five replicates of spiked marijuana samples at concentrations previously cited as QC1, QC2, and QC3 controls. For the determination of the inter-assay precision, three replicates of spiked marijuana samples were analyzed in five different days.

Accuracy—Accuracy was defined as the difference between the mean result and the expected result, expressed as a percentage. In this study, accuracy was determined through the calculation of average QC (1, 2, and 3) concentrations in triplicate. The relative area values were converted into herbicide concentration values through the respective linear regression equations obtained from the analytical curves. Accuracy was considered as the percentage of systematic error expressed by sample bias (the difference between the real and the obtained values) (8).

Extraction Recovery—For the recovery study regarding the two developed methods, two marijuana sample groups containing different concentrations were analyzed for each method. The extraction recovery was calculated by comparing the mean area response of extracted samples (spiked before extraction) to that of unextracted samples (spiked after extraction) at each QC's concentration level.

Results and Discussion

As previously described, marijuana is the most consumed illegal drug and seized by the police all over the world. Brazil is the fourth in the rank of seizures (3.0% of total), and 8.8% of the population has already used the drug (1,2).

Capillary electrophoresis is a useful tool to forensic analyses, mainly for determination of high polarity compounds such as paraquat, glyphosate, and AMPA. Among other features, the presence of charges in the molecular structure allows developing simple and fast separation methods (9).

Two methods that use capillary electrophoresis were developed for the present study: one in free solution with direct UV detection, aiming at paraquat determination in *Cannabis* samples, and another using capillary electrophoresis in free solution with indirect detection through UV/VIS for glyphosate and AMPA

detection. The use of indirect detection was necessary because glyphosate and AMPA do not present chromophoric or fluorophoric groups. These molecules present zwitterionic behavior, with many possible ionizations. As consequence, the state of charges and its mobility can be changed according to the pH. In pH = 4.6, glyphosate presents a positive charge and two negative ones (as shown in Fig. 2), and AMPA presents a positive charge and another negative, so they become totally appropriate to be analyzed (see Fig. 3). On its turn, paraquat presents two amine groups positively charged in a broad pH range, so it is fully appropriate to be analyzed through CE (Fig. 1).

The electrophoretic conditions employed were adequate, leading to a good separation and a satisfactory resolution of the peaks of interest. The time employed for each analysis was 2.5 min for paraquat and 9 min for glyphosate and AMPA.

According to the optimized electrophoretic and extraction parameters, the number of possible interferences in both methods is not significant. No interference was found in the analyses. The herbicide DQ, analyzed as a possible interference, can be differentiated from the other compounds, paraquat and ethyl paraquat, through the absolute migration time (2.19 min paraquat and 2.21 min DQ), relative migration time (0.90 for paraquat and 0.91 for DQ), and UV spectrum (obtained in diode-array detector), as depicted in Fig. 4. No analyzed sample presented contamination by diquat.

For glyphosate and AMPA analyses, no interference was detected. These herbicides were differentiated from possible interference peaks through the absolute migration time (3.87 min glyphosate and 7.72 min AMPA) and relative migration time (0.77 for glyphosate and 1.55 for AMPA).

The develop methods were homoscedastic (average of variances was below the F factor), so the concentrations of analytes were calculated from the simple linear regression equation established by the least squares method. The dynamic range was from 5 to 1000 $\mu\text{g/g}$ (paraquat) and 10 to 1000 $\mu\text{g/g}$ (glyphosate and AMPA). The correlation coefficients (r) obtained in the analytical curves of the methods for all analytes were >0.999 . The confidence parameters of the validated methods were summarized in Table 1.

TABLE 1—Method validation regarding limit of detection (LOD), limit of quantitation (LOQ), linearity, precision, accuracy, and recovery for the determination of paraquat, glyphosate, and AMPA in marijuana by capillary electrophoresis.

Parameter	Paraquat	Glyphosate	AMPA
LOD ($\mu\text{g/g}$)	0.8	8	8
LOQ ($\mu\text{g/g}$)	5	10	10
Linearity			
Slope	0.0204	0.0181	0.0277
Intercept	0.0141	0.0487	0.0728
Coefficient of correlation	0.9999	0.9998	0.9997
Regression error	0.0145	0.0153	0.0281
Intra-day precision (% RSD)			
QC1	4.8	2.7	5.5
QC2	1.9	0.9	1.2
QC3	1.6	1.1	1.9
Inter-day precision (% RSD)			
QC1	8.2	5.3	7.9
QC2	5.2	3.6	4.3
QC3	2.1	4.4	6.8
Accuracy (%)			
QC1	90.2	95.8	90.8
QC2	97.8	98.7	92.4
QC3	98.5	99.2	94.9
Recovery (%)			
QC1	92.1	89.4	91.3
QC2	96.5	92.1	93.5
QC3	97.7	93.7	92.9

QC 1 = 50 $\mu\text{g/g}$; QC 2 = 400 $\mu\text{g/g}$; QC 3 = 800 $\mu\text{g/g}$; % RSD, relative standard deviation.

A broad range of concentration was evaluated for the linearity study. Shown inside the dynamic interval of the methods, this range was established by the method for paraquat determination based on the concentrations found in marijuana samples contaminated in the 70's (10–13). Meanwhile, these concentrations were found when the purpose of application was culture eradication. As eradication programs through paraquat no longer exist, the purpose of using this herbicide is unknown, although it might be used to improve *Cannabis* cultivation or act in the drug drying process. It might also be used for removing occasional weeds that could impair the cultivation or it simply could stem from

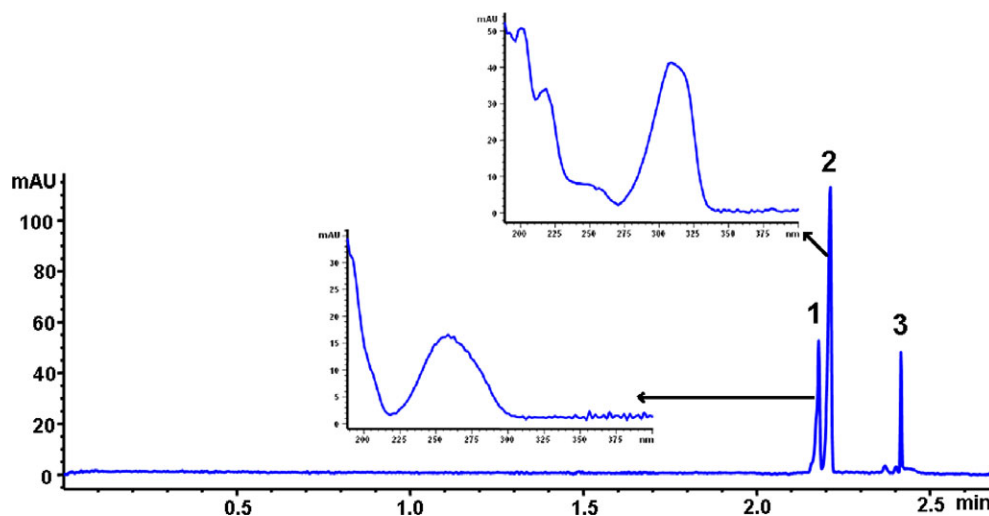


FIG. 4—Electropherogram and UV spectra of marijuana blank sample spiked with (1) paraquat 50 $\mu\text{g/g}$, (2) diquat 100 $\mu\text{g/g}$, and (3) ethyl paraquat 50 $\mu\text{g/g}$ (internal standard). Electrophoretic conditions: fused silica capillary 75 μm i.d., 48.5 $\times$ 40 cm (effective); electrolyte: phosphate buffer 50 mmol/L (pH 2.50); 21 kV, 21°C, 195 nm; hydrodynamic injection: 50 mbar/10 sec.

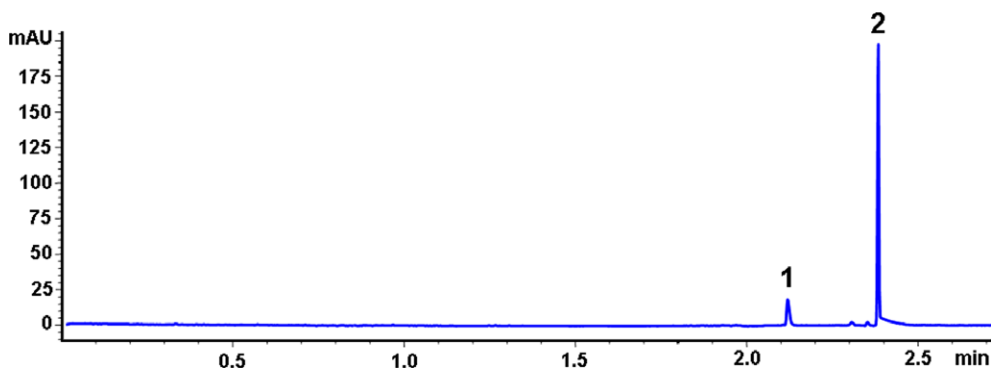


FIG. 5—Electropherogram of marijuana sample, with positive result for presence of paraquat. Peaks: (1) paraquat (2.7 $\mu\text{g/g}$); (2) ethyl paraquat (IS). Electrophoretic conditions see Fig. 4.

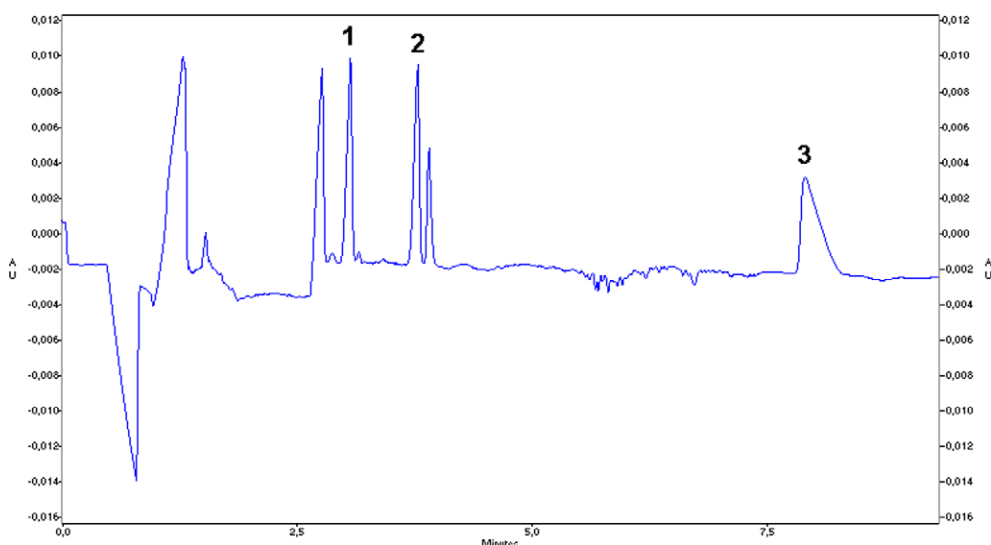


FIG. 6—Electropherogram of marijuana sample, with positive result for presence of glyphosate and AMPA. Peaks: (1) glyphosate (55.8 $\mu\text{g/g}$); (2) glutamic acid (IS); (3) AMPA (38.3 $\mu\text{g/g}$). Electrophoretic conditions: fused silica capillary 75 μm i.d., 37 $\times$ 30 cm (effective); electrolyte: 10 mmol/L galic acid, 6 mmol/L TRIS, and 0.1 mmol/L CTAB (pH = 4.7); -20 kV, 25°C, indirect detection at 254 nm; hydrodynamic injection: 3 psi/5 sec.

TABLE 2—Paraquat concentrations in seized marijuana samples.

Sample No.	Total Seized Amount (kg)	Paraquat (mg/g)
1	1	0.38
2	2×10^3	1.06
3	12×10^{-3}	0.02
4	5×10^{-3}	0.15
5	47	25.1
6	12×10^{-3}	3.47
7	25×10^{-3}	0.67
8	3×10^3	2.26
9	70×10^{-3}	0.15
10	4×10^{-3}	0.04
11	12	0.01
12	7×10^{-3}	0.45

the contamination of the soil where other cultures used to be previously grown.

Regarding the method for glyphosate and AMPA determination, the concentration range was in accordance with glyphosate expected concentrations in several cultures, in addition to the residual findings in soil (14,15). However, these concentrations are unexplainable for applications in illegal substances that do not require use-related care, because the research on other herbicides such as glyphosate had not been performed yet.

TABLE 3—Glyphosate and AMPA concentrations in seized marijuana samples.

Sample No.	Total Seized Amount (kg)	Glyphosate (mg/g)	AMPA (mg/g)
13	1	0.15	ND
14	5	0.75	ND
15	10	0.55	0.36

ND, not detected.

The limit of detection and quantification obtained in this study can be considered as satisfactory (Table 1) when the extraction technique, the range of concentration and the detection system employed are taken into account, in addition to the fact that this methodology is completely new to forensic applications.

The inter-assay and intra-assay precision calculated as coefficient of variation presented adequate results. They were always below 10% for both studies. The use of ethyl paraquat as an appropriate internal standard (structurally related to paraquat) and glutamic acid (structurally related to glyphosate and AMPA), which allows the respective participations in every analytical stage of the developed methods contribute to the obtainment of these factors (in addition to other factors).

Whenever available, the accuracy of an analytical method must be evaluated using samples whose analyte concentration is guaranteed by normative institutions or entities responsible for laboratory quality assurance programs (certified samples). When this type of material is not available, as in the case of certified *Cannabis* samples with paraquat, glyphosate, and AMPA, it is acceptable that the accuracy of the analytical method is evaluated through spiking blank samples, comparing the values found with the nominal added concentration. The accuracy of an analytical method should be between $\pm 85\%$, so the proposed method can be considered adequate (8,16).

The acceptable recovery intervals for the analysis of residues generally vary from 70% and 120% with a precision of up to $\pm 20\%$. However, depending on the analytical complexity of the sample, this value can vary from 50% to 120%, with a precision rate of up to $\pm 15\%$ (16). The recovery values obtained in this work comply with the acceptable values, always above 85%.

Regarding analytes stability, it was observed that if marijuana spiked with paraquat was stored without air contact, there was no significant decay of herbicide concentration. However, when samples underwent the same conditions, but with direct contact with air, it was observed fungi contamination and significant decreased of paraquat levels. The paraquat degradation in soil by different fungi in soil was reported previously (15, 17). Authors depict that the main fungi responsible for this degradation are the ones belonging to the *Aspergillus* and *Penicillium* genders, the same identified by our research group (Rafael Lanaro (RL), Silvia Oliveira Santos Cazenave (SOSC) unpublished data). A similar degradation pattern was observed with glyphosate (with formation of AMPA).

The real samples submitted to the developed method were seized by São Paulo state police in a year (total amount = 20 tons). Among the 130 analyzed samples, 12 were contaminated by paraquat in different concentrations ranging from 0.01 to 25.1 mg/g (mean = 2.81 mg/g), which correspond to over five tons of herbicide-contaminated marijuana. The drug was seized as cigarettes, shopping bags containing the drug on the loose, small packages, and compressed packages.

Among the 130 seized samples, only three were contaminated by herbicide glyphosate, and one of them presented concomitant contamination by AMPA. The concentrations were 0.15, 0.75, and 0.55 mg/g for glyphosate and 0.36 mg/g for AMPA. These results show that two samples with glyphosate had probably been exposed to the herbicide not long ago because AMPA could not be detected. On the other hand, another sample had been exposed to glyphosate for a longer period of time as it was possible to detect the presence of its own breakdown product.

The electropherograms of real samples submitted to the proposed methods are presented in Figs 5 and 6. Tables 2 and 3 depict the total seized amount and herbicides concentrations in analyzed samples (note: all results are expressed in mg of herbicide per gram of sample).

Conclusion

The results presented herein demonstrate that the validated methods are simple, rapid, and executable in forensic toxicology laboratories. The validation parameters results are compatible with the values available in the literature. Comparative data for the methodology proposed for glyphosate and AMPA determination in marijuana samples are not available, but the ones obtained by the present research proved to be satisfactory when compared to the values deemed as residual and available in food cultures.

Paraquat concentrations in marijuana samples showed huge variation. It did not present degradation under the influences imposed by the postharvest and storage periods. Glyphosate and AMPA contaminations in *Cannabis sativa* L samples were not demonstrated in the previously chosen samples to be analyzed because both analytes undergo intense degradation under the influences imposed by the postharvest and storage periods in room temperature. However, it was detected in three samples of the 30 samples added to the study for glyphosate and AMPA determination, with low and varied concentrations. Finally, experimental studies are suggested to evaluate the real risks posed by paraquat, glyphosate, and AMPA, in found concentrations, for marijuana users.

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